FetalAI – Fetal Health Prediction System

1. Introduction

FetalAI is a machine learning-based system designed to predict the health condition of an unborn baby during pregnancy. Monitoring fetal health is very important to ensure safe delivery and avoid complications. This project uses medical data and predictive modeling to classify fetal health into three categories: Normal, Suspect, and Pathological. The system helps doctors and pregnant women in early detection of risks and improves decision-making in healthcare.

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2. Problem Statement

Traditional fetal monitoring requires continuous hospital visits and manual observation, which can be time-consuming and sometimes inefficient. There is a need for an automated system that can quickly analyze fetal health parameters and provide accurate predictions to assist healthcare professionals.

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3. Objective

- To develop a system that predicts fetal health using machine learning
- To classify fetal condition into Normal, Suspect, and Pathological
- To provide a simple web interface for user interaction
- To assist in early detection of fetal risks

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4. Scope of the Project

- Can be used by doctors for quick analysis
- Helps pregnant women monitor baby health
- Can be extended for real-time monitoring systems
- Useful in healthcare and medical research fields

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5. Technologies Used

- **Programming Language:** Python
- **Framework:** Flask
- **Frontend:** HTML, CSS, Bootstrap
- **Machine Learning:** Scikit-learn
- **Model Storage:** Pickle (.pkl file)

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6. System Architecture

1. User enters input data through UI
2. Data is sent to Flask backend
3. Backend processes input and sends it to ML model
4. Model predicts fetal health condition
5. Result is displayed on UI

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7. Dataset & Features

The model uses **21 input parameters**, including:

- Baseline fetal heart rate
- Uterine contractions
- Fetal movements
- Accelerations and decelerations
- Short-term and long-term variability
- Histogram-based statistical features

These features help in analyzing fetal condition accurately.

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8. Methodology

- Data collection and preprocessing
- Exploratory Data Analysis (EDA)
- Model training using multiple algorithms
- Model evaluation using performance metrics
- Selection of best-performing model
- Model saving using .pkl file
- Integration with Flask web application

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9. Working of the System

- User opens the web application
- Enters all required medical inputs
- Clicks on Predict button
- Model processes the input data
- Output is displayed as:
    - **Normal** (Healthy)
    - **Suspect** (Needs monitoring)
    - **Pathological** (High risk)

10. Results

The system successfully predicts fetal health based on given inputs. It provides quick and reliable results, which can help in early diagnosis and preventive care.

11. Advantages

- Fast and accurate prediction
- Reduces need for frequent hospital visits
- Easy to use interface
- Supports early risk detection

12. Limitations

- Depends on quality of input data
- Not a replacement for medical diagnosis
- Requires proper model training for accuracy

13. Future Enhancements

- Integration with real-time monitoring devices
- Mobile application development
- Improved accuracy with advanced models
- Cloud-based deployment

14. Conclusion

FetalAI demonstrates how machine learning can be applied in healthcare to predict fetal health conditions. It helps in early detection of risks and supports better decision-making for doctors and patients. This project shows the potential of AI in improving pregnancy care and medical analysis.

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15. References

- Machine Learning documentation
- Healthcare datasets (fetal health dataset)
- Python and Flask official documentation